

Global nonlinear simulation of frequency sweeping of hot electron interchange modes in a dipole-confined plasma

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A nonlinear model based on the vorticity equation is developed for describing the low-frequency hot electron interchange instability in a dipole-confined plasma containing an energetic population of trapped electrons. We introduce a source-driven, nonlinear simulation of the flute-like electrostatic response to a steep radial gradient of energetic electrons deeply trapped in a magnetic dipole. The global simulation with a sustained hot electron source shows the phase-space dynamics of energetic electrons and the increasing frequency of the saturated instability.

I. INTRODUCTION

The closed-field-line dipole configuration is relevant to the study of planetary magnetospheres [1–5], magnetic fusion confinement [6–9], and magnetically confined non-neutral plasmas [10–13]. The hot electron interchange (HEI) instability, first described by Krall [14], plays a key role in the radial transport of energetic electrons in the magnetic mirror and dipole configurations [15]. These energetic particle instabilities represent interesting and important physics. The HEI has been extensively studied in laboratory experiments, particularly in CTX [1] and LDX [16–18]. In these dipole experiments, the hot electrons are equatorially trapped near the microwave cyclotron resonance, producing steep radial gradients in the hot electron number density profiles. Consequently, a population of hot electrons trapped in the dipole field can excite low-frequency, flute-type electrostatic fluctuations through drift-resonant wave-particle interactions, leading to cross-field transport and the formation of complex phase-space structures [19, 20].

Observations in CTX demonstrate that the HEI instability driven by hot electrons exhibits rich nonlinear behavior, including frequency sweeping and the spontaneous formation of phase-space holes [1, 21]. Frequency sweeping can be suppressed by applying radio frequency fields, which is attributed to the destruction of phase-space holes by localized velocity-space diffusion [22, 23]. Furthermore, a fully self-consistent model for the nonlinear evolution of the HEI has been developed [1, 21]. In this model, the potential fluctuations are computed from the Poisson equation, which gives an elliptic equation for the time derivative of the potential in terms of the $\mathbf{E} \times \mathbf{B}$ convection of the fluctuating charge density, the divergence of the nonaxisymmetric ion polarization currents, and the magnetic drift of the hot electrons [1, 21].

However, this approach requires evolving the ion continuity equation and using iterative methods for the elliptic equation, resulting in significant computational cost [21]. The previous simulations of HEI based on solving the Poisson equation have demonstrated the inward motion of phase-space holes [19] and the rising frequency of the instability [1, 21].

For the experiments in CTX, both the observations and the numerical simulations appear to be explained by a generalized nonlinear theory developed by Berk, Breizman and co-workers [24, 25]. For a driven mode near marginal stability, $\gamma_L \sim \gamma_d$, where γ_L is the linear growth rate of the instability due to resonant particle effects and γ_d is the dissipation rate due to nonresonant effects. In this theory, waves with frequency sweeping are generated as the instability grows to a large amplitude. During the saturation stage, the wave frequencies continue to change as energy extracted from the resonant particles is balanced by nonresonant dissipation.

In this work, we present a self-consistent, nonlinear model based on the vorticity equation for the evolution of a quasineutral plasma consisting of cold ions and a mixture of hot and cold electrons confined by a dipole magnetic field. The vorticity model explores the flute-like electrostatic response to a steep radial gradient of hot electrons confined in a dipole geometry. The nonlinear simulations describe the self-consistent dynamics of the hot electrons and the growth and saturation of the low-frequency HEI. This description is relatively simple because the low-frequency electrostatic instabilities are flute-like, and the dynamics of the deeply trapped hot electrons are two-dimensional. A source term for hot electron production adds to the distribution evolution equation, which allows the hot electron profile to evolve self-consistently. Previously, a conserving source term has been included in the ion continuity equation for simulating quasi-steady driven interchange turbulence [26]. Both the observations and the previous simulations show that the fluctuations induced by the HEI are dominated by the azimuthal mode number $m = 1$ [1, 21]. Thus we

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focus on the $m = 1$ mode in the present simulation.

This paper is organized as follows. Section II presents a brief description of the basic equations in dipole geometry. The vorticity equation for the quasineutrality is introduced in Sec. III. The linear stability of the HEI is described in Sec. IV. In Sec. V the fully nonlinear simulations of the HEI are presented. In the final section, we summarize key conclusions.

II. BASIC EQUATIONS

In the following, we summarize a self-consistent, nonlinear model for the low-frequency HEI in dipole magnetic flux coordinates. The model describes a plasma comprising cold ions and a mix of cold and hot electrons. The instability of interest is flute-like and electrostatic with a frequency much higher than the cold electron drift frequency. The axisymmetric dipole field is $\mathbf{B} = \nabla\varphi \times \nabla\psi = \nabla\chi$, where χ is the coordinate along the field line, ψ represents the magnetic flux and φ is the azimuthal angle.

Charged particle motion in a dipole magnetic field is characterized by three types of frequencies (cyclotron, ω_c , bounce, ω_b , and drift, ω_d). When the waves are flute-like with drift resonances, $\omega \approx m\omega_d \ll \omega_b \ll \omega_c$, radial motion can occur while preserving both μ and J [15]. For an electrostatic perturbation in the low-frequency limit ($\omega \ll \omega_b$), the dynamics of a population of trapped hot electrons is described by the bounce-averaged distribution, $F(\mu, J, \psi, \varphi, t)$ [27]. When the particle distribution is expressed in terms of adiabatic constants, (μ, J) , the collisionless evolution of F is [1, 27]

$$\frac{\partial F}{\partial t} + \frac{\partial}{\partial \varphi} [(\omega_d + v_\phi) F] + \frac{\partial}{\partial \psi} (v_\psi F) = 0, \quad (1)$$

where the $\mathbf{E} \times \mathbf{B}$ drift velocities are

$$v_\phi = -\frac{\partial \Phi}{\partial \psi}, \quad v_\psi = \frac{\partial \Phi}{\partial \varphi}. \quad (2)$$

Since $B \propto \psi^3$ in a magnetic dipole, the azimuthal magnetic drift frequency of hot electrons is

$$\omega_d = \frac{\mu}{e} \frac{\partial B}{\partial \psi} = \frac{3\mu B}{e\psi}. \quad (3)$$

Since μ and J are conserved, hot electron dynamics is described in a two-dimensional (ψ, φ) canonical phase space [1, 27].

The number of particles in a tube per unit magnetic flux is defined by

$$N(\psi, \varphi) = \int \frac{d\chi}{B^2} \iiint d^3v F. \quad (4)$$

The velocity integral can be written as [27]

$$\iiint d^3v = \frac{2\pi B}{m^2} \sum_{+-} \iint \frac{d\mu dE}{|v_\parallel|}, \quad (5)$$

where the kinetic energy $E = m(v_\perp^2 + v_\parallel^2)/2$ and magnetic moment $\mu = mv_\perp^2/(2B)$, and $\sum_{+-}$ represents the sum over the positive and negative parallel velocities. Since F is the bounce-averaged distribution, Eq. (4) can be written

$$N = \frac{2\pi}{m^2} \iint d\mu dE \left(\sum_{+-} \int \frac{d\chi}{B} \frac{1}{|v_\parallel|} \right) F, \quad (6)$$

where the sum over the field-line integral defines the full bounce period:

$$\tau_b = \oint \frac{dl}{|v_\parallel|} = \sum_{+-} \int \frac{d\chi}{B} \frac{1}{|v_\parallel|}. \quad (7)$$

Thus we find

$$N = \frac{2\pi}{m^2} \iint \tau_b dE d\mu F. \quad (8)$$

Consider the longitudinal invariant

$$J = \oint dl |v_\parallel| = \oint dl \sqrt{\frac{2}{m}(E - \mu B)}, \quad (9)$$

where $B(l)$ is the magnitude of the magnetic field along the field line. Changing the integration variable from E to J at constant μ , we use

$$dJ = \left(\frac{\partial J}{\partial E} \right)_\mu dE = \frac{\tau_b}{m} dE. \quad (10)$$

Then Eq. (8) becomes

$$N = \frac{2\pi}{m} \iint dJ d\mu F. \quad (11)$$

For deeply trapped hot electrons with $J = 0$, the distribution becomes $F = F(\mu, \psi, \varphi, t) \delta(J)$. Integrating over J further, we obtain

$$N_h = \int d\mu F \quad (12)$$

Here, the constant $2\pi/m$ in Eq. (11) has been absorbed into the definition of F .

The field-line averaged Poisson equation is [1]

$$\int \frac{d\chi}{B^2} \nabla_\perp^2 \Phi = -4\pi e \int \frac{d\chi}{B^2} (n_i - n_e) = -4\pi e (N_i - N_e), \quad (13)$$

where N_i and N_e are the number of ions and electrons within the flux tube. In dipole magnetic flux coordinates, the flux-tube integral of the divergence of a quantity $\mathbf{A}$

is given by [27]

$$\int \nabla \cdot \mathbf{A} \frac{d\chi}{B^2} = \frac{\partial}{\partial \psi} \int \nabla \psi \cdot \mathbf{A} \frac{d\chi}{B^2} + \frac{\partial}{\partial \varphi} \int \nabla \varphi \cdot \mathbf{A} \frac{d\chi}{B^2}. \quad (14)$$

Axial symmetry implies $|\nabla \varphi| = 1/R$ and $|\nabla \psi| = RB$ with R the major radius from the dipole axis. Using Eq. (14) and $\nabla_\perp \Phi = (\partial \Phi / \partial \psi) \nabla \psi + (\partial \Phi / \partial \varphi) \nabla \varphi$, we find

$$\int \frac{d\chi}{B^2} \nabla^2 \Phi = \frac{\partial}{\partial \psi} \int \frac{d\chi}{B^2} R^2 B^2 \frac{\partial \Phi}{\partial \psi} + \frac{\partial}{\partial \varphi} \int \frac{d\chi}{B^2} \frac{1}{R^2} \frac{\partial \Phi}{\partial \varphi}. \quad (15)$$

For a point dipole, the geometric integrals along the field line can be obtained (see Appendix). Then Eq. (13) becomes [1]

$$\frac{\partial}{\partial \varphi} \left(2 \frac{M}{\psi^2} \frac{\partial \Phi}{\partial \varphi} \right) + \frac{\partial}{\partial \psi} \left(4M \frac{\partial \Phi}{\partial \psi} \right) = -4\pi e (N_i - N_e), \quad (16)$$

where the electrostatic potential has been assumed to be constant along the dipole field line.

The evolution of cold ion density is described by the continuity equation $\partial n / \partial t + \nabla \cdot (n \mathbf{v}) = 0$, where $\mathbf{v}$ is approximated by the sum of $\mathbf{E} \times \mathbf{B}$ and polarization drifts. Use of Eq. (14) yields the field-line averaged $\mathbf{E} \times \mathbf{B}$ convection term

$$\int \frac{d\chi}{B^2} \nabla \cdot (n \mathbf{v}_E) = \frac{\partial}{\partial \psi} (N v_\psi) + \frac{\partial}{\partial \varphi} (N v_\varphi). \quad (17)$$

With Eq. (14) we obtain the flux-tube integrated polarization drift term

$$\begin{aligned} & \int \frac{d\chi}{B^2} \nabla \cdot \left(\frac{n M_i}{B^2} \frac{\partial}{\partial t} \nabla \Phi \right) \\ &= \frac{\partial}{\partial \varphi} \left(N_i \Upsilon_\varphi \frac{\partial}{\partial \varphi} \dot{\Phi} \right) + \frac{\partial}{\partial \psi} \left(N_i \Upsilon_\psi \frac{\partial}{\partial \psi} \dot{\Phi} \right), \end{aligned} \quad (18)$$

where the dot denotes the time derivative, i.e., $\dot{\Phi} \equiv \partial \Phi / \partial t$. For simplicity, the nonlinear advective part of the polarization drift has been neglected. The density-weighted field line averages are defined as [1, 27, 28]

$$\Upsilon_\varphi = \frac{1}{N_i} \int \frac{d\chi}{B^2} \frac{n M_i}{B^2} \frac{1}{R^2}, \quad \Upsilon_\psi = \frac{1}{N_i} \int \frac{d\chi}{B^2} \frac{n M_i}{B^2} R^2 B^2. \quad (19)$$

Then the continuity equation for cold ions becomes [1]

$$\begin{aligned} & \frac{\partial N_i}{\partial t} + \frac{\partial}{\partial \psi} (N_i v_\psi) + \frac{\partial}{\partial \varphi} (N_i v_\varphi) \\ &= \frac{\partial}{\partial \varphi} (N_i u_\varphi) + \frac{\partial}{\partial \psi} (N_i u_\psi), \end{aligned} \quad (20)$$

where we have only included the linear part of the polarization drift,

$$u_\varphi = \Upsilon_\varphi \frac{\partial^2 \Phi}{\partial \varphi \partial t}, \quad u_\psi = \Upsilon_\psi \frac{\partial^2 \Phi}{\partial \psi \partial t}. \quad (21)$$

III. DERIVATION OF VORTICITY EQUATION

It is convenient to nondimensionalize the model equations. We denote $x = \psi / \psi_0$ as the normalized magnetic flux and $y = \varphi$ as the azimuthal angle, where $\psi = M / L$ represents the magnetic flux through the equatorial plane with an equatorial radius L and the dipole moment M . The magnetic field and density are normalized to their values at an equatorial reference radius L_0 . The magnetic moment of hot electrons μ is normalized by a reference value μ_0 . The electrostatic potential Φ is normalized to T_{e0}/e with $T_{e0} = 3\mu_0 B_0$. The frequency is normalized to the characteristic magnetic drift frequency of hot electrons,

$$\omega_{d0} = \frac{3\mu_0 B_0}{e\psi_0}. \quad (22)$$

Then Eq. (1) for the evolution of hot electron distribution $F(x, y, \mu, t)$ is normalized to

$$\frac{\partial F}{\partial t} + \frac{\partial}{\partial y} [(\hat{\omega}_d + v_y)F] + \frac{\partial}{\partial x} (v_x F) = 0, \quad (23)$$

where

$$\hat{\omega}_d = \frac{\omega_d}{\omega_{d0}} = \mu x^2, \quad (24)$$

$$v_y = -\frac{\partial \Phi}{\partial x}, \quad v_x = \frac{\partial \Phi}{\partial y} \quad (25)$$

Integrating Eq. (23) over μ yields the hot electron continuity equation:

$$\frac{\partial}{\partial t} N_h + \frac{\partial}{\partial y} (v_y N_h) + \int d\mu \hat{\omega}_d F + \frac{\partial}{\partial x} (v_x N_h) = 0. \quad (26)$$

Cold electrons with $\mu \simeq 0$ execute radial and azimuthal $\mathbf{E} \times \mathbf{B}$ motion, according to

$$\frac{\partial}{\partial t} N_c + \frac{\partial}{\partial y} (v_y N_c) + \frac{\partial}{\partial x} (v_x N_c) = 0. \quad (27)$$

The total electron density is $N_e = N_c + N_h$. Adding Eqs. (26) and (27) gives the electron continuity equation

$$\dot{N}_e + \frac{\partial}{\partial y} (v_y N_e) + \frac{\partial}{\partial x} (v_x N_e) + \frac{\partial}{\partial y} \int d\mu \hat{\omega}_d F = 0. \quad (28)$$

The cold ion evolution equation (20) is normalized into

$$\dot{N}_i + \frac{\partial}{\partial y} (v_y N_i) + \frac{\partial}{\partial x} (v_x N_i) \simeq \frac{\omega_{d0}}{\omega_{ci0}} \left(\frac{\partial}{\partial y} u_y + \frac{\partial}{\partial x} u_x \right), \quad (29)$$

where the N_i in the polarization drift term has been taken to be the equilibrium density with $N_i \simeq 1$. For a point dipole [27], $\Upsilon_\psi \simeq 0.7/x^2$ and $\Upsilon_\varphi \simeq 0.6/x^4$. For simplicity, we further assume $\partial u_x / \partial x \simeq \Upsilon_\psi \partial^2 \dot{\Phi} / \partial x^2$.

The dimensionless form of Poisson equation (16) is

$$\frac{2}{x^2} \frac{\partial^2}{\partial y^2} \Phi + 4 \frac{\partial^2}{\partial x^2} \Phi = 0.91 \left(\frac{L_0}{\lambda_D} \right)^2 (N_e - N_i), \quad (30)$$

where $\lambda_D^2 = T_{e0}/4\pi N_0 e^2$ is the Debye length. For typical experimental values $(L_0/\lambda_D)^2 \sim 10^3 \gg 1$ [21], the right hand side of Eq. (30) implies the quasineutrality condition $N_e = N_i$, or equivalently

$$\dot{N}_e - \dot{N}_i = 0 \quad (31)$$

With Eqs. (28) and (29) the quasineutrality condition (31) yields the vorticity equation:

$$\frac{\partial}{\partial y} \int d\mu \hat{\omega}_d F + \frac{1}{\hat{\omega}_{ci0}} \left(\frac{0.6}{x^4} \frac{\partial^2}{\partial y^2} + \frac{0.7}{x^2} \frac{\partial^2}{\partial x^2} \right) \Phi = 0, \quad (32)$$

where $\hat{\omega}_{ci0} \equiv \omega_{ci0}/\omega_{d0}$. This vorticity equation couples the hot electron magnetic drift with the cold ion polarization drift, providing a description of the HEI modes.

IV. LINEAR HEI MODES

Because of axial symmetry in a dipole configuration, the perturbations are expressed in Fourier series, $\Phi(x, y, t) = \sum_m \Phi_m(x) \exp(imy - i\omega t)$, where ω is the complex frequency and m is the azimuthal mode number. Then the linearized equation (23) gives the drift-resonant fluctuations,

$$\tilde{F} = \frac{m\Phi}{\omega - m\hat{\omega}_d} \frac{\partial F_0}{\partial x}, \quad (33)$$

where F_0 is the equilibrium distribution. The perturbed hot electron distribution is driven by the potential perturbations and the radial gradients of background density profiles. When the electrostatic potential consists of azimuthally propagating waves, the drift resonance condition is

$$\omega = m\hat{\omega}_d = m\mu x^2, \quad (34)$$

which relates a particle's energy μ to a resonant flux surface $x = \sqrt{\omega/m\mu}$.

The hot electron equilibrium distribution takes the form [1]

$$F_0(x, \mu) = \alpha N_{i0} G(\mu) \delta(J) \quad (35)$$

where $\alpha(x) = N_h/N_{i0}$ is the fractional density of hot electrons, $G(\mu)$ is the normalized magnetic moment distribution satisfying $\int G(\mu) d\mu = 1$, and $\delta(J)$ indicates that electrons are deeply trapped in the equatorial plane ($J = 0$). Taking the Fourier transform of Eq. (32) at

$x \simeq 1$, we obtain

$$m \int d\mu \hat{\omega}_d \tilde{F} + \frac{1}{\hat{\omega}_{ci0}} m_{\perp}^2 \omega \Phi = 0, \quad (36)$$

where $m_{\perp}^2 = 0.6m^2 + 0.7k_x^2$ and the wavenumber k_x represents the radial mode width.

Substitution of Eqs. (33) and (35) into Eq. (36) yields

$$m^2 \frac{\partial(N_{i0}\alpha)}{\partial x} \int d\mu \frac{\hat{\omega}_d G(\mu)}{\omega - m\hat{\omega}_d} \Phi + \frac{1}{\hat{\omega}_{ci0}} m_{\perp}^2 \omega \Phi = 0. \quad (37)$$

When $G(\mu') = \delta(\mu' - \mu)$, the hot electrons are mono-energetic. Then Eq. (37) gives the simplified dispersion relation [1]:

$$\omega^2 - m\hat{\omega}_d \omega + \Gamma_h^2 = 0, \quad (38)$$

where $\hat{\omega}_d = \mu x^2$ is the hot electron drift frequency and $\Gamma_h^2 \equiv (m^2/m_{\perp}^2) \hat{\omega}_{ci0} \hat{\omega}_d \partial(N_{i0}\alpha)/\partial x$ is the hot electron interchange drive. The solution to the quadratic equation (38) gives the complex frequency in the form

$$\omega = \frac{m\hat{\omega}_d}{2} + \frac{1}{2} \sqrt{m^2 \hat{\omega}_d^2 - 4\Gamma_h^2}. \quad (39)$$

The mode has a real frequency $\omega = m\omega_d/2$ and the HEI instability occurs when the hot electron density gradients exceed the threshold,

$$\frac{\partial(N_{i0}\alpha)}{\partial x} > \frac{\omega_d}{\omega_{ci0}} \frac{m_{\perp}^2}{4}. \quad (40)$$

The HEI modes are more unstable with small perpendicular mode number $m_{\perp}$ (low values of m and broad radial mode structures of the electrostatic potential, $k_x \simeq 0$). The drive for the instability is the radial phase-space density, not the energy distribution. Thus the energetic electrons with a Maxwellian distribution and overly peaked radial profiles can also drive the HEI.

V. NONLINEAR SIMULATION OF THE HEI

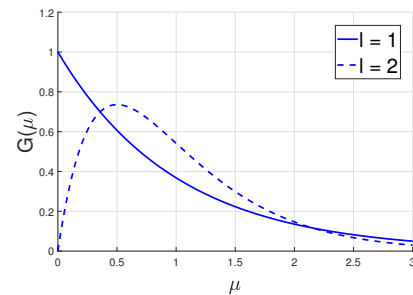


FIG. 1: Energy distribution $G(\mu)$ of hot electrons.

The normalized magnetic moment distribution of the

hot electrons is modeled by a gamma distribution

$$G(\mu) = \frac{l^l}{\Gamma(l)} \mu^{l-1} \exp(-l\mu) \quad (41)$$

with $\Gamma(l)$ the gamma function. When $l = 1$, Eq. (41) represents a Maxwellian distribution of hot electrons. The normalized μ space is discretized with the normalization for the distribution $\sum_i G(\mu_i) \Delta\mu = 1$ where $\mu_i = i \Delta\mu$ with $i = 1, \dots, 5$ and an interval of $\Delta\mu = 0.25$. Figure 1 shows the distribution. For each μ value, the Vlasov equation for the full F is evolved without separation of equilibrium and perturbation, (23)

$$\frac{\partial F}{\partial t} + \frac{\partial}{\partial y} [(\hat{\omega}_d + v_y)F] + \frac{\partial}{\partial x} (v_x F) = S, \quad (42)$$

where $S(x, \mu) = S_0(x)G(\mu)$ represents the hot electron source term. The drive term of the vorticity equation $\int d\mu \hat{\omega}_d F$ is calculated by $\sum_i \mu_i x^2 F(\mu_i) \Delta\mu$, once the F has been advanced in time. Fourier transforming the vorticity equation (32) in y gives

$$\left(-m^2 \frac{0.6}{x^4} + \frac{0.7}{x^2} \frac{\partial^2}{\partial x^2} \right) \dot{\Phi}_m(x) = w_m(x), \quad (43)$$

where $w = \hat{\omega}_{ci0} \int d\mu \hat{\omega}_d \partial F / \partial y$. This equation can be inverted to obtain $\dot{\Phi}_m(x)$ using a standard tridiagonal solver, and then an inverse transform in y gives $\dot{\Phi}(x, y)$. To model non-resonant dissipation, we introduce a diffusion term in the potential update:

$$\Phi(t + \Delta t) = \Phi(t) + \Delta t \dot{\Phi} + \nu \Delta t \nabla^2 \Phi, \quad (44)$$

where ν is the diffusion coefficient and $\nabla^2 \Phi = \partial^2 \Phi / \partial x^2 + \partial^2 \Phi / \partial y^2$.

We have developed an explicit, finite-difference code for solving Eqs. (42)-(44) on uniform, rectangular grids in the (x, y) plane. This code is a modification of our earlier code used for the study of interchange modes [28–31]. The nonlinearities arise from the convective term in Eq. (42), which takes a conservative form. The gradients of the flux are evaluated with fourth-order accuracy in the grid spacing and a velocity-dependent hyperviscosity is introduced for the convective term [32]. Equations (42) and (44) are advanced using a second-order trapezoidal leapfrog algorithm [32]. The simulation domain is radially global with the outer wall at $x_{min} = 0.3$ and the inner wall at $x_{max} = 1.5$. Note that the magnetic flux varies as the inverse of the equatorial radius, $x = \psi/\psi_0 \propto 1/L$. The radial boundaries are assumed to be conducting walls and therefore Dirichlet boundary conditions are imposed with $\Phi = 0$ and $F = 0.1$ at the outer and inner walls. Periodic boundary conditions are used for all quantities in the azimuthal direction for $y \in [0, 2\pi]$. The constant parameters used in the simulations are $\hat{\omega}_{ci0} = \omega_{ci0}/\omega_{d0} = 3.5$ [1], the diffusion coefficient $\nu = 0.002$, and the time step $\Delta t = 0.01$. The

hot electron source is radially localized with a Gaussian profile, which has a peak amplitude $S_0 = 0.005$ and is centered at $x = 1$.

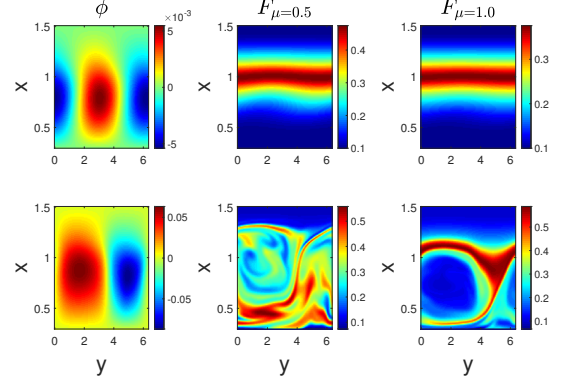


FIG. 2: Snapshot of structures of Φ and $F(\mu)$ with $\mu = 0.5$ and $\mu = 1$ at the linear (top) and nonlinear (bottom) stages.

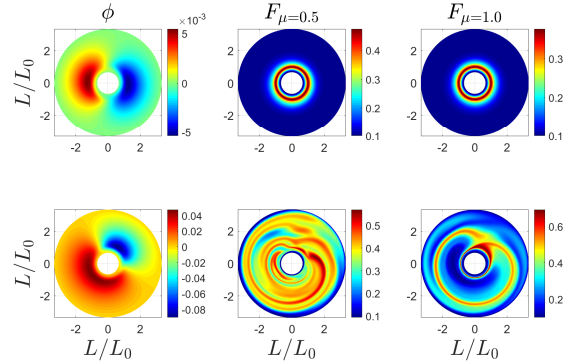


FIG. 3: Snapshot of structures in the equatorial plane with polar coordinates at the linear (top) and nonlinear (bottom) stages.

We now turn to the simulation of Eqs. (42)-(44), which shows the self-consistent evolution of the HEI in dipole geometry. Initially, small perturbations are added to the potential with $m = 1$ in the y direction. The initial hot electron profiles are uniform in y (i.e., axisymmetric) and radially localized at the source location, as shown in Fig. 2 (top). Here the magnitude of $F \propto \alpha$ essentially represents the fractional density of hot electrons.

When the radial gradients of F become sufficiently steep, the HEI modes are excited in the outer region of density peak where $dF/dx > 0$, as shown in Fig. 2 (top). The potential perturbations develop small perpendicular mode number $m_\perp$ with broad radial structures, as expected from the local linear theory. The perturbations grow exponentially in amplitude and have azimuthal propagation in the electron magnetic drift direction (positive y , i.e., counterclockwise). Here the magnetic field points out of the plane in Fig. 2. The linear

growth rate is obtained from an exponential fit to the slope of $\log \Phi_{rms}(t)$, where $\Phi_{rms}^2 = \langle \Phi^2 \rangle_{x,y}$ is the square of perturbations averaged over the entire x, y domain. Time evolution of the potential gives a linear growth rate of $\gamma_L \sim 0.16$ for the HEI. Figure 3 shows the structures in the equatorial plane. Here Fig. 3 is obtained by transforming from rectangular grids in Fig. 2 to polar coordinates with the equatorial radius $L/L_0 = \psi_0/\psi = 1/x$.

In the nonlinear phase, the potential structure has a broad radial extent and circulates azimuthally in the direction of the electron magnetic drift. The potential fluctuations have considerable harmonic structure in the azimuthal direction with a dominant mode number $m = 1$, as shown in Fig. 2 (bottom). As the intensity of perturbations grows to sufficiently large amplitudes comparable to the background equilibrium of F , the first instability burst rapidly expels the hot electrons radially. At saturation of the instability, the phase-space structures of F depend on the particle's energy, as shown in Fig. 2 (bottom). For electrons with low energy $\mu \sim 0.5$, the instability drives strong interchange mixing across the outer region. The turbulent mixing results in a broad spectrum of azimuthal mode numbers. In contrast, at a high value of $\mu \sim 1$, the phase-space structure is dominated by the mode number $m = 1$. The hot electrons circulate clockwise around the boundary of the positive potential peak, resulting in a large-scale, low-density structure, called a phase-space “hole”, in the outer region. This coherent structure of F is nearly rigid and propagates together with the potential structure in the positive y direction.

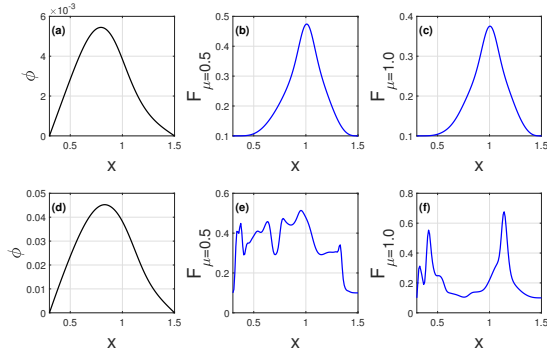


FIG. 4: Snapshot of profiles of Φ and $F(\mu)$ with $\mu = 0.5$ and $\mu = 1$ at a given azimuthal angle $y = \pi$ in the linear (top) and nonlinear (bottom) phases.

In the linear phase, as shown in Fig. 4(a), the potential profiles exhibit a broad peak near $x \simeq 0.8$ with wide radial extent. The equilibrium distribution is peaked at $x \simeq 1$ where the hot electron source is located, as seen in Figs. 4(b) and 4(c). Thus the distribution has maximum positive gradients near $x \simeq 0.8$, which drives the HEI modes. In the nonlinear stage, the potential has a broad, rigid radial structure, implying the formation of large-scale convective cells, as seen in Fig. 4(d). With the diffusion term in Eq. (44), the dissipation rate of the

mode is defined as

$$-\gamma_d = \nu \frac{1}{\Phi} \frac{\partial^2 \Phi}{\partial x^2}, \quad (45)$$

With the radial profiles of the potential, the second derivatives $\partial^2 \Phi / \partial x^2 < 0$ can be calculated and then the dissipation rate can be estimated as $\gamma_d \sim 0.16/3$ for a diffusion coefficient of $\nu = 0.002$. This estimate indicates weakly unstable modes with $\gamma_L - \gamma_d < \gamma_L$ for the linear growth rate of $\gamma_L \sim 0.16$.

The phase-space structures created by the HEI vary with the particle's energy. At low $\mu \sim 0.5$, strong radial transport results in marginal stability characterized by $\partial F / \partial x \approx 0$ in the outer unstable region, as seen in Fig. 4(e). For electrons with higher energy $\mu \sim 1$, the density peak originally at the source location moves further inward so that the location of steep positive gradients of F shifts to the plasma center with larger values of $x \simeq 1.1$, where particles resonantly interact with weakly unstable discrete modes. The density peak is also formed at the outer edge $x \simeq 0.4$, and the width between the peaks of the distribution indicates the radial extent of the phase-space “hole”, as shown in Fig. 4(f). Note that the radial gradient of F at the outer edge does not result in drift-resonant fluctuations according to Eq. (33), since the potential amplitudes almost vanish near the wall boundary.

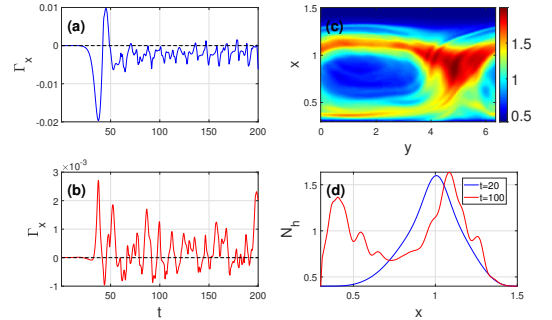


FIG. 5: Time evolution of the averaged radial flux at (a) $x = 0.8$ and (b) $x = 1.2$. (c) Spatial structure and (d) radial profile of N_h at a particular instant of time.

Figure 5 shows the collisionless radial transport of hot electrons induced by the instability. The averaged radial flux is defined by $\Gamma_x = \langle N_h v_x \rangle_y$ where the bracket $\langle f \rangle_y = \int f dy / 2\pi$ denotes the azimuthal average and $v_x = \partial \Phi / \partial y$ represents the radial $\mathbf{E} \times \mathbf{B}$ motion. Here, positive (negative) values of Γ_x indicate inward (outward) transport. The hot electron number density is obtained by $N_h = \sum_i F(\mu_i) \Delta \mu$. At the linear stage, the radial flux grows exponentially with the largest amplitude occurring at $t \simeq 40$, leading to the first instability burst. As the instability enters the saturated phase, the flux amplitude reduces to a lower level and exhibits bursty fluctuations, as seen in Figs. 5(a) and 5(b). The

time averaged transport of hot electrons is dominated by the inward flux with $\Gamma_x > 0$ in the inner region $x > 1$, leading to the inward shift of the number density peak from $x = 1$ to $x \approx 1.1$, as seen in Figs. 5(c) and 5(d). In contrast, the average particle flux is radially outward with $\Gamma_x < 0$ in the outer region $x < 1$, resulting in the build-up of the number density near the outer boundary $x \approx 0.4$. The inward and outward transport of particles are sustained by continuous particle source localized near the plasma center $x \approx 1$.

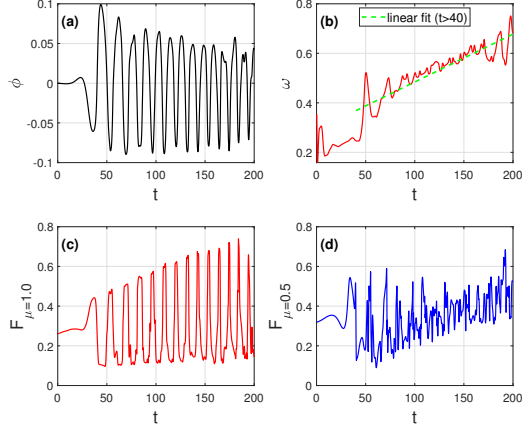


FIG. 6: Time evolution of (a) potential, (b) mode frequency for $m = 1$, (c) F for $\mu = 1$, and (d) F for $\mu = 0.5$ at a fixed point $x = 0.8$ and $y = \pi$.

Figure 6(a) shows the time evolution of the potential at a fixed point. The potential grows exponentially during the linear stage ($t < 40$) and then saturates with a slowly decreasing amplitude due to the dissipation. The oscillations of the potential result from the azimuthal circulation of convective cells in the nonlinear stage, as shown in Fig. 2 (bottom). The mode frequency is defined by the phase factor $\exp(-i \int \omega dt)$. Fourier transform of the potential $\Phi(x, y, t)$ in y yields $\Phi_m(x, t) = |\Phi_m| \exp(i\theta_m)$ with the phase $\theta_m(x, t) \equiv \arctan(\Im \Phi_m / \Re \Phi_m)$. With the phase $\theta_m = -\int \omega dt$, the mode frequency is obtained from the time derivative of the phase [33],

$$\omega_m(x, t) = -\frac{\partial}{\partial t} \theta_m(x, t). \quad (46)$$

The unwrapped phase $\theta(t)$ changes linearly in time during the linear stage ($t < 40$) and then becomes curved with an increasingly negative slope in the nonlinear stage, indicating the rising frequency of the waves. Thus the mode frequency is nearly constant at the linear stage with $\omega \approx m\omega_d/2 \simeq 0.25$ for $m = 1$, $\mu = 1$, and $x \simeq 0.7$, as expected from the local linear theory. Note that the linear growth rate is less than the mode frequency. At the first instability burst ($t \simeq 50$), the mode frequency jumps to a higher value and then slowly increases in time, as shown in Fig. 6(b). The chirping rate $\partial\omega/\partial t \simeq 0.002$

is obtained from a linear fit during the nonlinear stage, implying the azimuthal acceleration of the rigid potential structure. The magnetic drift frequency in a dipole field scales as $\omega_d \propto x^2$. As hot electrons with higher energy $\mu \sim 1$ move inward in a magnetic dipole, as seen in Fig. 4(f), their drift frequency increases. Thus the mode frequency increases according to the drift resonance condition Eq. (34).

In the nonlinear stage, the time evolution of the distribution with higher energy $\mu = 1$ is quasi-periodic, implying a frequency spectrum with coherent peaks, as shown in Fig. 6(c). The sharp drop in F indicates the passing of phase-space holes, which corresponds to the crest of the potential oscillations. This correlation can also be seen in the spatial structures of the potential and particle distribution in Fig. 2 (bottom). Meanwhile, the amplitude of F gradually increases due to the sustained hot electron source. In contrast, the nonlinear dynamics of low energy electrons with $\mu = 0.5$ is very turbulent with a broad frequency spectrum, as shown in Fig. 6(d).

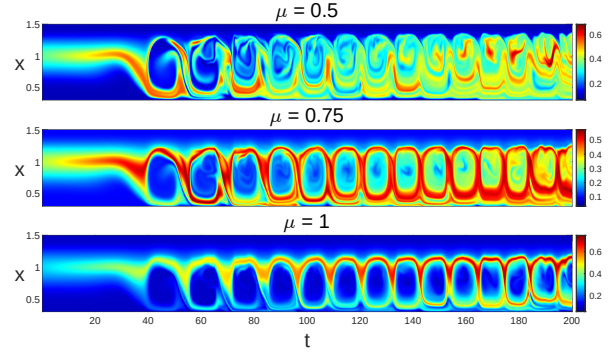


FIG. 7: Time evolution of the radial profile of F at a given $y = \pi$ distributed over different values of μ .

Figure 7 illustrates the whole process of phase-space evolution. As the number density increases due to the hot electron source, the HEI is eventually triggered by the radial gradients of F . As predicted in the linear theory Eq. (40), the HEI modes are more unstable at low hot-electron energy (since $\omega_d \propto \mu$) and high fractional density α , as seen in Fig. 7. After $t > 40$, the phase-space holes develop for electrons with higher energy $\mu \simeq 1.0$. These coherent structures are quasi-periodic in time. The initial mode frequency is low. As the amplitude of the instability saturates, the time interval between the repeated hole structures slowly reduces with a period $T \simeq 10$ at the late nonlinear stage, which corresponds to a frequency $\omega = 2\pi/T \simeq 0.6$ in Fig. 6(c). This value is consistent with the mode frequency from the local linear theory, $\omega = m\omega_d/2 = m\mu x^2/2 \simeq 0.6$ with $m = 1$, $\mu = 1$, and $x \simeq 1.1$ at the location of the positive gradient of F in Fig. 4(f). In contrast, the dynamics of low energy electrons with $\mu \simeq 0.5$ are characterized by interchange mixing and flattening of radial profiles across the outer region.

VI. CONCLUSION

The vorticity model governing the hot electron interchange instability has been derived for a dipole-confined plasma consisting of equatorially trapped hot electrons. The nonlinear simulations based on the vorticity equation demonstrate the hot electron phase-space evolution and radial transport induced by the low-frequency HEI. The global simulations with a sustained hot electron source show that the creation and dynamical evolution of phase-space structures are essential to the nonlinear saturation of the instability and the frequency sweeping of the modes in a dipole configuration. The vorticity model developed in this work provides an efficient tool for understanding energetic particle transport and nonlinear wave-particle interaction in dipole geometry.

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Appendix A: Geometric integrals for a point dipole

For a point dipole in spherical coordinates [1]

$$B = \frac{M}{r^3} \sqrt{1 + 3 \cos^2 \theta}, \quad \chi = \frac{M \cos \theta}{r^2}, \quad (\text{A1})$$

where M is the magnetic moment of the dipole. A dipole field line equation is given by $r = L \sin^2 \theta$. In the derivation of the flux-tube averaged Poisson equation, we have the following two geometric integrals:

$$I_1 = \int \frac{d\chi}{B^2} R^2 B^2, \quad I_2 = \int \frac{d\chi}{B^2} \frac{1}{R^2}, \quad (\text{A2})$$

where $R = r \sin \theta = L \sin^3 \theta$. Since

$$\frac{d\chi}{d\theta} = -\frac{M}{L^2} \frac{1 + 3 \cos^2 \theta}{\sin^5 \theta} \quad (\text{A3})$$

we obtain

$$I_1 = M \int_0^\pi \sin \theta (1 + 3 \cos^2 \theta) d\theta = 4M. \quad (\text{A4})$$

$$I_2 = \frac{L^2}{M} \int_0^\pi \sin \theta d\theta = 2 \frac{L^2}{M} = \frac{2M}{\psi^2}, \quad (\text{A5})$$

where we have used $\psi = M/L$ to label a flux surface and a particle's equatorial radial position [1].

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